



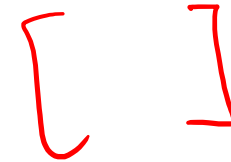
Linear Transformations

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Recap

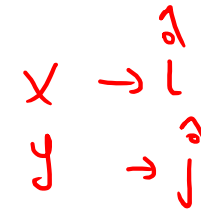
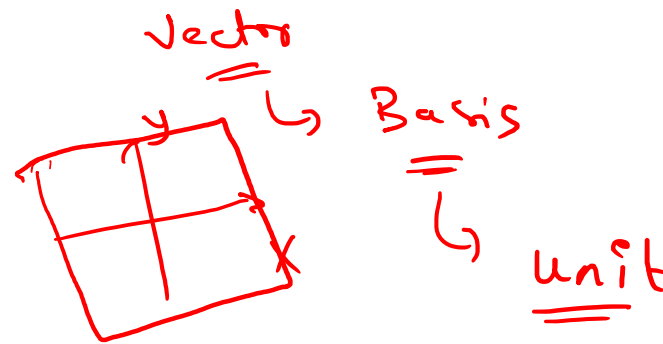
- Matrices

- Physical significance $\rightarrow$ info. matrices
- Operations of Matrix $\rightarrow$ image
- Special Matrices $\checkmark$ $\rightarrow$ pixel



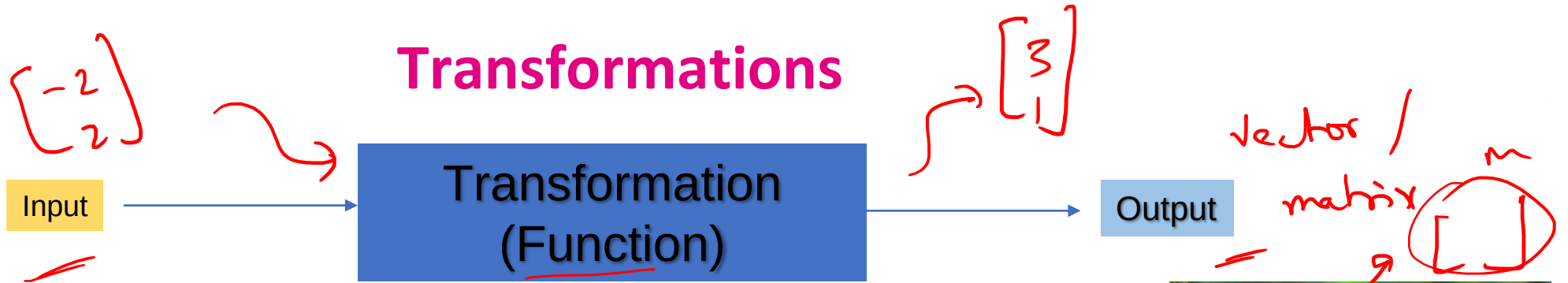
- Vectors

- Significance $\rightarrow$ CS
- Linear Combination $\checkmark$
- Linearly dependent
- Basis and Unit vectors



Transformations

vector / matrix

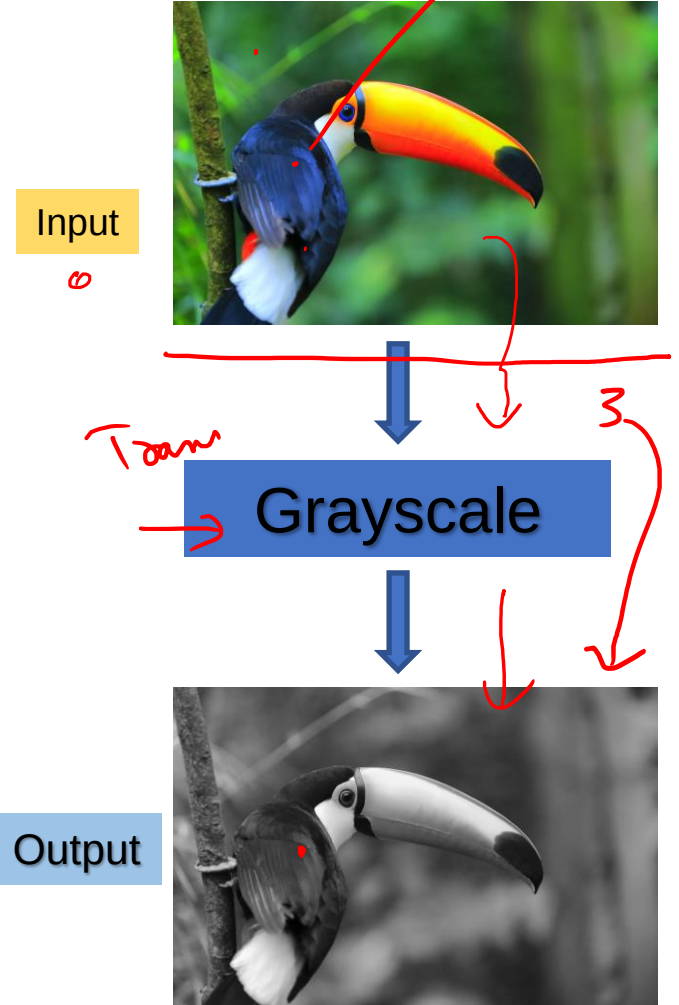
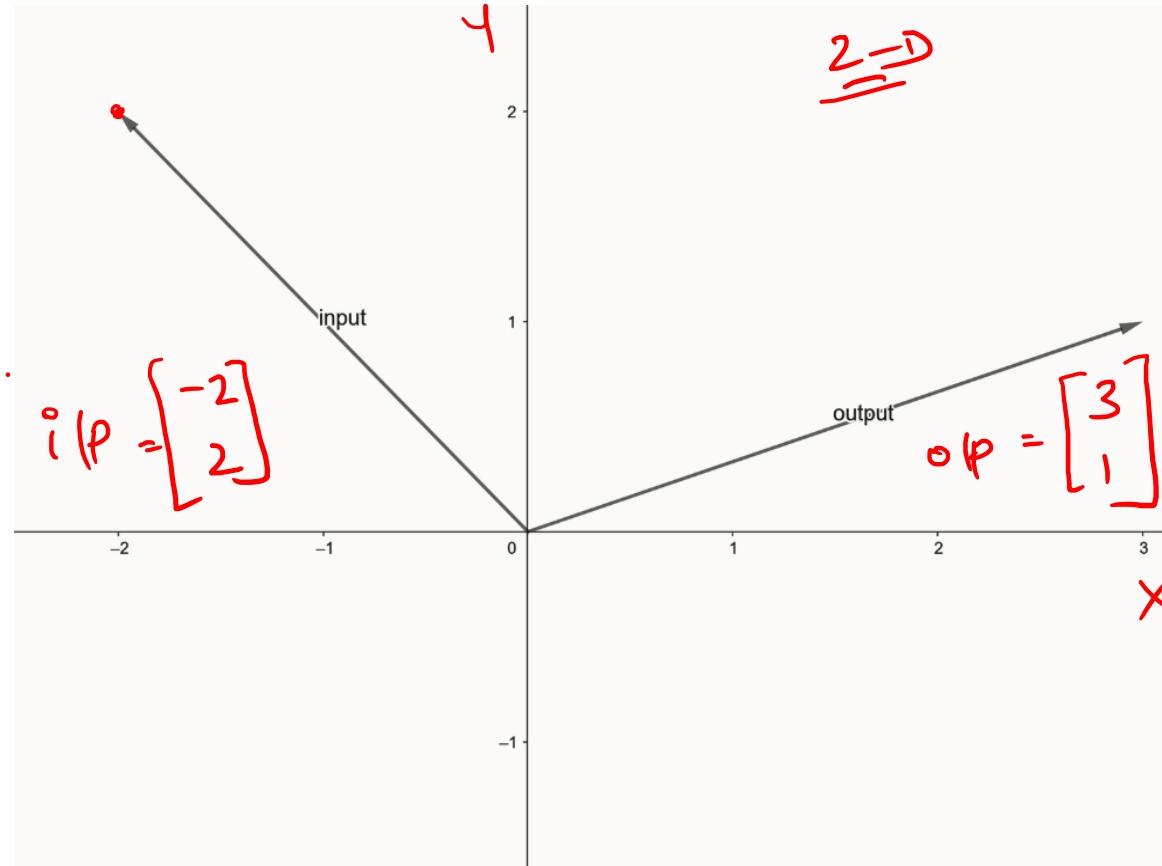


$$f(x) = x^2$$

① → 1
② → 4

A transformation can be thought of as some movement in the vectors.

It takes some vector as an input and gives us another vector (just like a function)



Linear Transformation

Mathematically speaking: A transformation is linear if it satisfies Additivity and Homogeneity:

Suppose a transformation is defined as

$$T: V \rightarrow W$$

$\begin{matrix} \text{input} & & \text{output} \\ \downarrow & & \uparrow \end{matrix}$

Additivity:

$$T(u + v) = Tu + Tv$$

Homogeneity:

$$T(\alpha u) = \alpha T(u)$$

* Linear transformations are sometimes also known as linear map

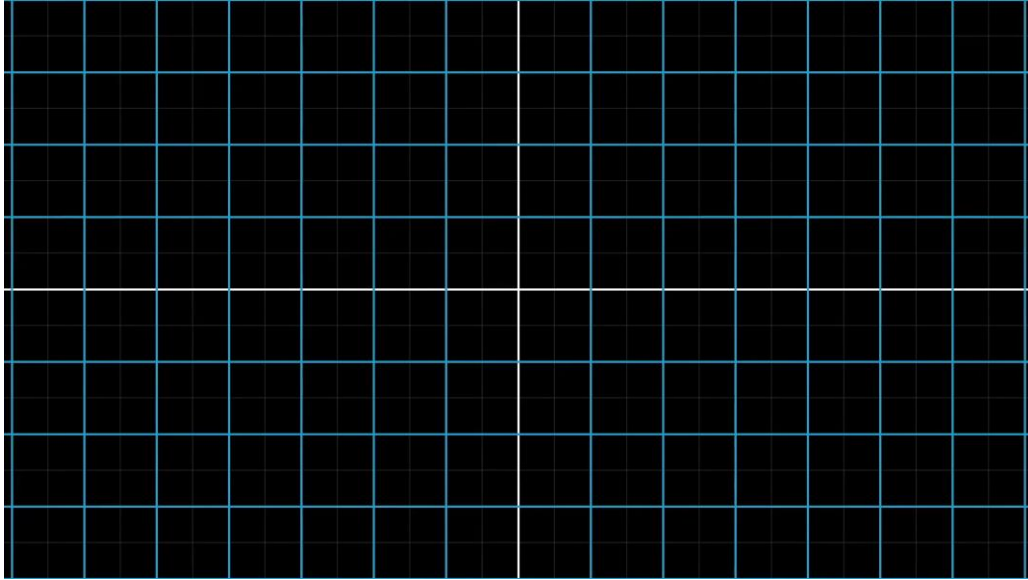
$$\begin{array}{ccc}
 T(\vec{v}) & & T(\vec{u} + \vec{v}) \\
 \{ T(\vec{u}) & & \downarrow \text{addition} \\
 T(\vec{v}) & & \downarrow \\
 & & T(\vec{u})
 \end{array}$$

$$\begin{array}{c}
 \alpha \cdot \vec{u} \\
 \downarrow \text{scalar}
 \end{array}$$

Linear Transformation

→ i/p () o/p

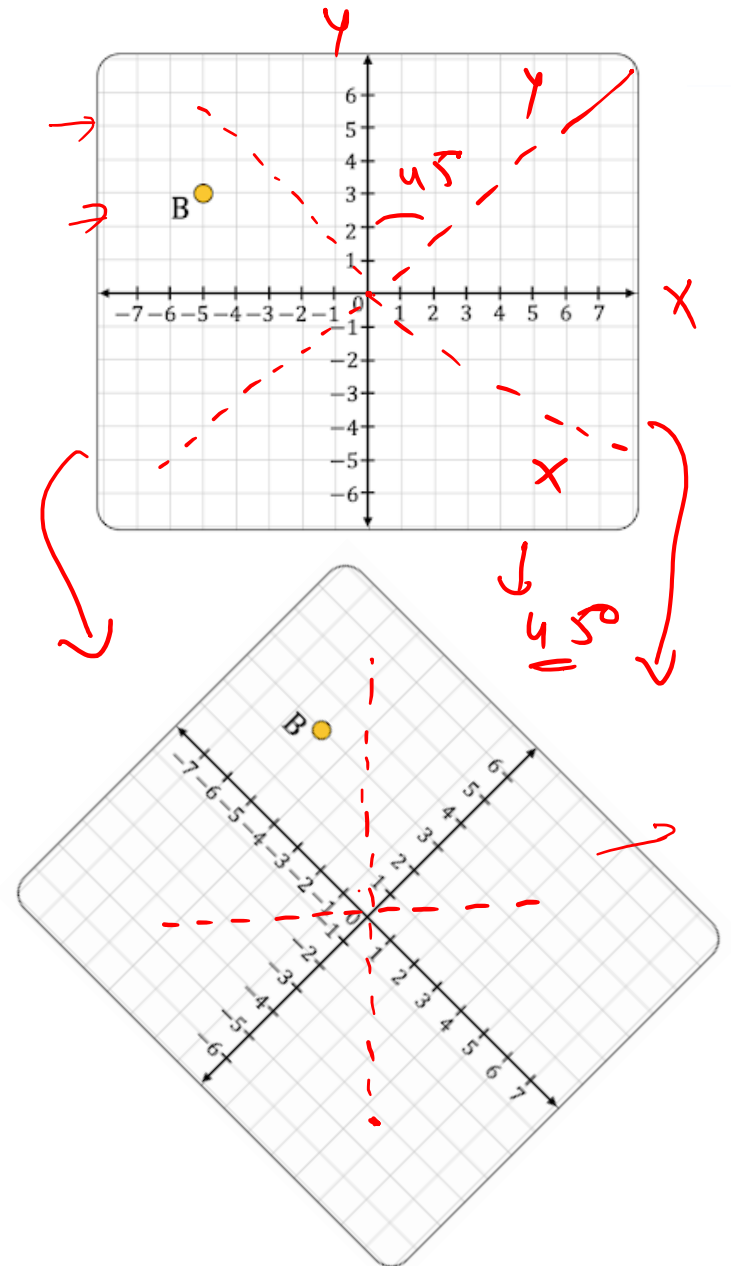
What sense does a transformation makes in physical terms:



Source: 3blue 1 brown

Physically speaking: A transformation is linear if:

- 1. All grid lines after transformation must remain straight lines and not get curved.
- 2. The origin must remain fixed after applying the transformation.



Linear Transformation

Summary: A linear transformation can be thought of as a movement in the reference coordinate axes system keeping all the grid lines parallel and evenly spaced.

Some linear transformations are pretty straight forward:

Ex. Rotation of axes about the origin.



Linear Transformation

2D (x, y) span
 $\hat{i}, \hat{j}$

For linear transformation the most important thing is to know is what happens to the basis (unit) vectors after transformation?

Ex. $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \rightarrow T$

$\vec{v} \rightarrow T(\vec{v})$

back $\rightarrow \hat{i}, \hat{j}$

$T \rightarrow \hat{i}_{\text{new}} = \begin{bmatrix} 4 \\ -2 \end{bmatrix} \quad \hat{j}_{\text{new}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$T(\vec{v}) = 3 \begin{bmatrix} 4 \\ -2 \end{bmatrix} + 4 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} 12 \\ -6 \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 16 \\ -6 \end{bmatrix} = \vec{w} = 16\hat{i} - 6\hat{j}$

$T(\vec{v})$

Generalized form

$\vec{v} = x\hat{i} + y\hat{j} \approx \begin{bmatrix} x \\ y \end{bmatrix}$

$L(\hat{i}) = \begin{bmatrix} i_1 \\ j_1 \end{bmatrix} \quad L(\hat{j}) = \begin{bmatrix} i_2 \\ j_2 \end{bmatrix}$

$\vec{v} \rightarrow L(\vec{v}) = x \begin{bmatrix} i_1 \\ j_1 \end{bmatrix} + y \begin{bmatrix} i_2 \\ j_2 \end{bmatrix}$

matrix transformation $= \begin{bmatrix} xi_1 + yi_2 \\ xj_1 + yj_2 \end{bmatrix}$

$L(\vec{v}) \rightarrow \begin{bmatrix} i_1 & i_2 \\ j_1 & j_2 \end{bmatrix}$

$L(\vec{v}) \cdot \vec{v} = \vec{w}$

transformation matrix

o/p after

Linear Transformation

Linear Transformation in 2-D space

$$L(\hat{i}) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, L(\hat{j}) = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} x \\ y \end{bmatrix}$$

transformation matrix: $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$
 $\vec{w}$

→ Shear transformation:

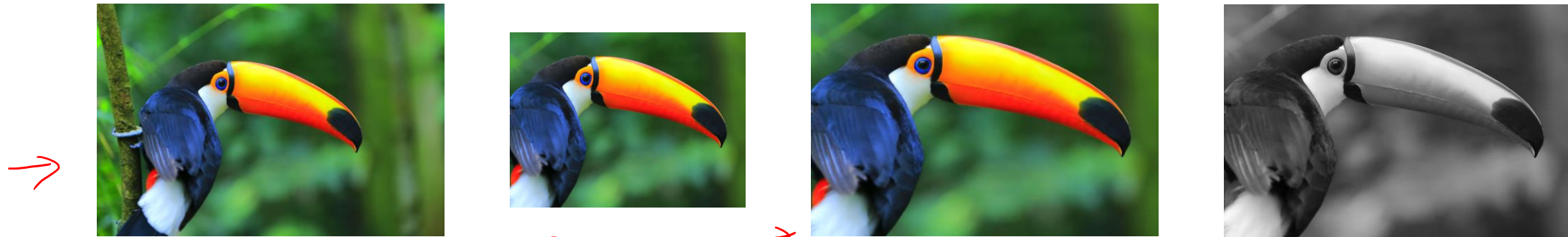
$\hat{i} \rightarrow$ remains fixed

$$L(\hat{i}) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, L(\hat{j}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$L(\text{Shear}) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Linear Transformation

In real-life scenario you will observe that we generally apply a series of transformations:



Image

Crop

Resize

Grayscale

Composition

$$L_1(\vec{v}) \rightarrow \textcircled{1}$$

$$L_2(\vec{v}) \rightarrow \textcircled{2}$$

$$L_2(L_1(\vec{v})) = \vec{w}$$

$$\text{Composition} = L_2 \cdot L_1$$

$$L_2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$L_1 = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

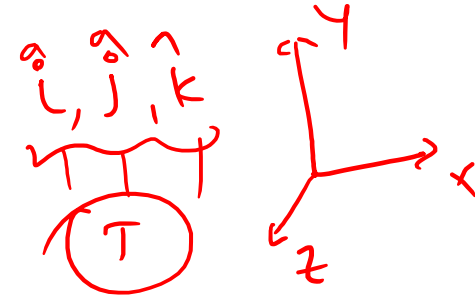
$$L_2 \cdot L_1 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bh & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

$$C \cdot V \begin{bmatrix} \cdot & \cdot \end{bmatrix} \begin{bmatrix} \cdot \\ \cdot \end{bmatrix} \rightarrow \vec{w}$$

function $\hookrightarrow f(g(n))$

Linear Transformation

Linear Transformation in 3-D space



transformation matrix $\rightarrow 3 \times 3$ matrix

$$L(\hat{i}), L(\hat{j}), L(\hat{k})$$

$\downarrow$
3-r

e.g. $\hat{i} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \hat{j} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \hat{k} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$$L(\vec{v}) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \vec{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\vec{w} = L(\vec{v}) \cdot \vec{v}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

n-D space

$$L(\vec{v}) = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \rightarrow$$

Linear Transformation

How to check if a given transformation is linear?

Linear

Additivity

Hom.

Ans. non-linear

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) \rightarrow \vec{w}$$

$$\begin{bmatrix} x-y \\ x+y \\ 3x \end{bmatrix}$$

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

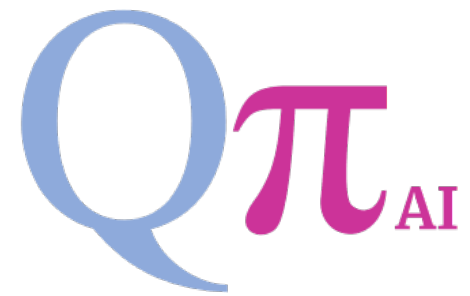
Add: $T(u+v) = T(u) + T(v)$
 Hom: $T(c \cdot u) = c \cdot T(u)$

$$T(u+v) = T\left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = T\left(\begin{bmatrix} u_1+v_1 \\ u_2+v_2 \end{bmatrix}\right) \Rightarrow \vec{w}_1 = \begin{bmatrix} u_1+v_1 - (u_2+v_2) \\ u_1+v_1 + (u_2+v_2) \\ 3(u_1+v_1) \end{bmatrix}$$

$$T(cu) = T\left(c \cdot \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}\right) = T\left(\begin{bmatrix} cu_1 \\ cu_2 \end{bmatrix}\right) \Rightarrow \vec{w}_1 = \begin{bmatrix} cu_1 - cu_2 \\ cu_1 + cu_2 \\ 3cu_1 \end{bmatrix}$$

$$c \cdot T(u) = c \cdot \begin{bmatrix} u_1 - u_2 \\ u_1 + u_2 \\ 3u_1 \end{bmatrix} = \begin{bmatrix} cu_1 - cu_2 \\ cu_1 + cu_2 \\ 3cu_1 \end{bmatrix} = \vec{w}_2$$

$\vec{w}_1 = \vec{w}_2$



Thank you!